

When comparing both methodologies, it becomes overwhelmingly clear why Linear Mixed Models (LMM) edge out traditional Repeated Measures ANOVA (RM-ANOVA) for longitudinal and repeated-measures datasets.

To see why, let's revisit our starting dataset: a sample of 10 patients across 4 repeated time points, with 3 total missing data points

Obs	Patient_ID	Group	Age	Week_0	Week_4	Week_8	Week_12
1	P001	Treatment	45	82.5	78.1	70.4	62.1
2	P002	Treatment	52	79.0	74.3	68.0	60.5
3	P003	Control	38	85.2	84.0	86.1	84.9
4	P004	Treatment	61	91.4	85.0	.	71.3
5	P005	Control	49	75.0	76.2	74.8	77.0
6	P006	Treatment	34	88.1	80.2	73.5	65.4
7	P007	Control	55	80.3	81.5	79.9	.
8	P008	Control	42	86.0	85.5	87.2	86.5
9	P009	Treatment	47	93.7	88.4	81.0	72.8
10	P010	Control	50	78.5	.	79.0	80.2

1. Attrition rate : RM-ANOVA acts like an inflexible, unforgiving boss. If an employee misses just one out of four scheduled check-ins, RM-ANOVA fires them on the spot via listwise deletion, completely throwing out their entire contribution.

LMM acts like a modern, sensible manager. It acknowledges life happens, retains the patient, and simply evaluates the data points they *did* provide. By keeping Patients 4, 7, and 10 in the fold, LMM saved our study from a massive 30% attrition penalty, maximizing our sample size and preserving observational value.

2. Model best fit: RM-ANOVA forces rigid, often unrealistic assumptions about how data points relate over time (demanding strict sphericity). LMM, on the other hand, lets us test and select tailored covariance structures, like Compound Symmetry (AIC = 128.2), that mirror the true underlying variance within patients across study visits.

3. Individual Baseline Isolation

LMM explicitly accounts for random subject-level intercepts and slopes. By isolating baseline individual variation, LMM strips out noise and vastly increases your statistical power to detect real, true clinical breakthroughs.

4. Fixed Effects & Statistical Power

While both models reached similar conclusions, LMM demonstrated far greater sensitivity and statistical power because it utilized every single available observation instead of tossing incomplete rows:

- Main Effect of Group: Both models confirmed no overall group difference, though LMM provides a cleaner estimate (LMM, $F = 1.75$ vs. RM-ANOVA, $F = 1.98$).
- Main Effect of Time: Both showed significant change over time, but LMM captured a much stronger signal (LMM, $F = 219.01$ vs. RM-ANOVA, $F = 153.45$) due to its optimized covariance structure and full sample retention.

- **Group × Time Interaction:** Both models detected the widening trajectory gap over time, but LMM demonstrated higher statistical precision (LMM, $F = 264.10$ vs. RM-ANOVA, $F = 183.06$) because LMM leveraged all 37 available data points across subjects rather than discarding incomplete cases

Overall, this project highlights why LMM is preferred over RM-ANOVA due to its ability to account for every data point rather than discarding incomplete cases. This increases the statistical power of the analysis and provides richer insights for the clinical trial. So the next time you have a boss who fairly assesses your performance based on all your hard work instead of writing you off over a single missing piece, tell them they're running on LMM energy.